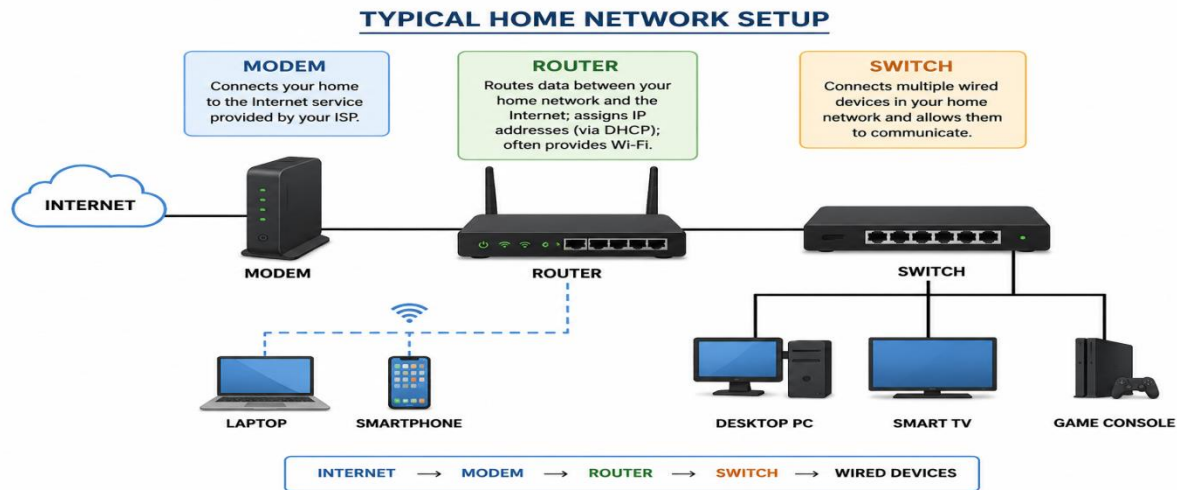


1. Draw a simple network diagram showing how a router, modem, and switch connect together in a typical home setup. Label each device and explain the role of each in one line on your diagram.



2. Log in to your home Wi-Fi router's admin panel (usually at 192.168.0.1 or 192.168.1.1) and find the section that shows connected devices. List the names or MAC addresses of at least 3 devices currently connected to your network.



Local Devices

Status	Connection Type	Device name	IP Address	MAC Address	IP Address Allocation	Lease Remaining	Li Ac Ti
Inactive	Wireless (5GHz)	Redmi-13-5G	192.168.1.65	ba:3f:3d:a1:1e:ed	DHCP	21 hours 44 min 18 sec	09/01 07:5 F
Active	Wireless (5GHz)	Redmi-Note-13-5G	192.168.1.64	ba:33:08:42:b0:cc	DHCP	23 hours 47 min 59 sec	10/01 07:1 A
Inactive	Wireless (2.4GHz)	Unknown_be:8b:1b:94:f3:0c	192.168.1.67	be:8b:1b:94:f3:0c	DHCP	11 hours 50 min 53 sec	11/01 12:4 A
Inactive	Wireless (2.4GHz)	krishna-s-S25-Ultra	192.168.1.75	16:3d:ee:53:5a:19	DHCP	7 hours 8 min 27 sec	13/01 05:0 A
Active	Wireless (2.4GHz)	ROHAN-s-S22	192.168.1.70	fe:b0:64:c7:24:78	DHCP	23 hours 22 min 27 sec	14/01 06:4 A
Active	Wireless (2.4GHz)	DESKTOP-SDT51D4	192.168.1.72	f8:2f:a8:c7:32:83	DHCP	18 hours 7 min 28 sec	15/01 04:3 F
Active	Wireless (2.4GHz)	Redmi-13-5G	192.168.1.69	ba:b6:3c:3b:43:b1	DHCP	23 hours 19 min 27 sec	15/01 07:1 F

3. Create a table comparing routers, modems, and switches. Include at least 4 points for each: main function, where it is used (home/office), whether it assigns IP addresses, and if it supports Wi-Fi.

Point	Router	Modem	Switch
Main function	Connects different networks and direct data between them.	Connects home/office network to the ISP's internet service.	Connects multiple devices within the same local network.
Where it is used	Homes and offices	Homes and offices	Mainly offices, businesses and larger networks
Assign IP addresses?	Yes, usually through DHCP	No normally does not assign local IP addresses	No normally does not assign IP addresses
Supports Wi-Fi?	Yes, many routers have built in Wi-Fi	Usually no, although some modem-router combination do.	No a standard switch uses Ethernet cables.

4. Imagine you are setting up a small gaming café. Choose and specify which devices (router, modem, switch) you would need, how many of each, and briefly explain the reason for each choice.

For a small gaming café, I would use the following network devices:

Device	Quantity	Reason
Router	1	Connect the café to the ISP and brings the internet connection into the café.
Modem	1	Connects the café network to the internet, assigns IP addresses using DHCP and

		provides basic network security.
Switch	2	Connects multiple gaming PCs and other wired devices together. Two switches can support more computers if one switch does not have enough ports.

Setup Example

- Internet/Modem/Routers/Switches/Gaming PCs.
- 1 modem – Receives the internet connection from the ISP.
- 1 router – Manages the network and provides Internet access to all devices.
- 2 switches – connects multiple gaming PCs and other wired devices together. Two switches can support more computers if one switch does not have enough ports.